ALGEBRA 2 · UNIT 4: POLYNOMIAL FUNCTIONS

Water Slide Designer

Design a smooth water slide with polynomial zeros, multiplicity, and end behavior.

CP1

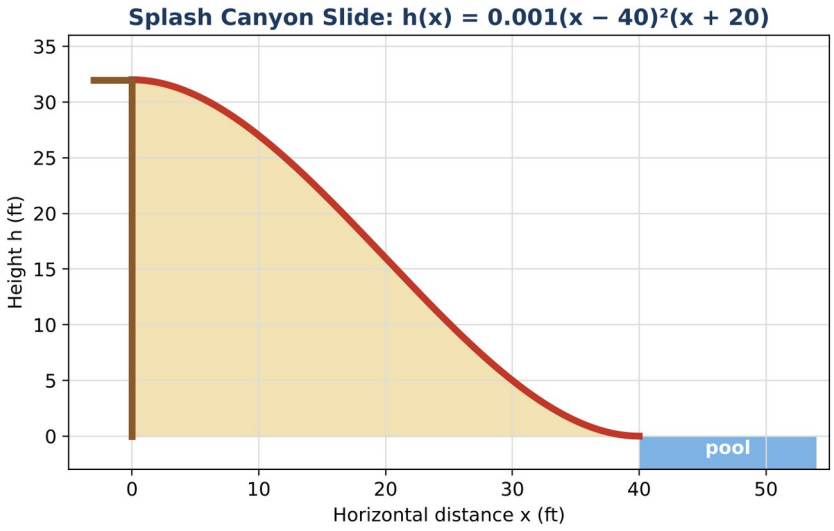
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Splash Canyon Water Park is building a new slide. The profile is $h(x) = 0.001(x - 40)^2(x + 20)$, where x is the horizontal distance in feet and h is the height in feet.

The double zero at x = 40 makes the slide level out smoothly at the pool. You will analyze this design and then create your own.



Learning Goals

- I can move between factored and standard forms of polynomials.
- I can use the Remainder, Factor, and Rational Root Theorems.
- I can describe end behavior and sketch graphs.

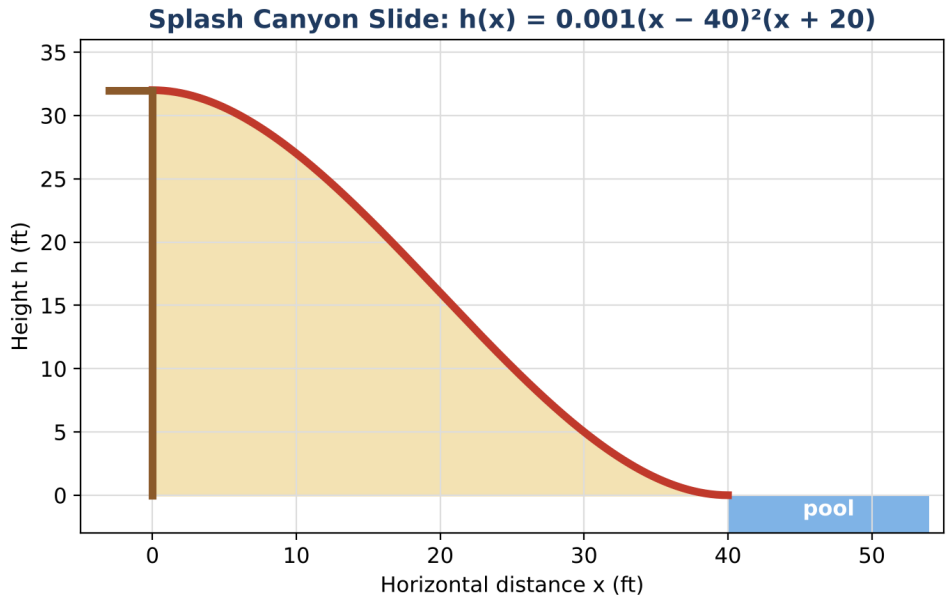
Project Timeline

When	What you will do
Day 1	Read the Slide
Day 2	Theorems at Work
Day 3	Design a Slide
Final	Slide Design Report + reflection

Part 1: Read the Slide

Day 1 · $h(x) = 0.001(x - 40)^2(x + 20)$

1. Expand to standard form. Check with x = 10.



2. Describe the end behavior of h (ignore the slide's domain).

3. Give the zeros, multiplicities, and the slide's domain and range.

4. Rule: the slide must be at least 10 ft high for the first 25 ft. Does it follow the rule?

Part 2: Theorems at Work

Day 2 · Use $p(x) = x^3 - 60x^2 + 32{,}000$.

5. Use the Remainder Theorem to find $p(30)$ with synthetic division. Then find $h(30)$.

6. List the possible rational roots of p according to the Rational Root Theorem (just describe the set). Then test $x = -20$.

7. Use the Factor Theorem and division to factor p completely.

Part 3: Design a Slide

Day 3 · Start height 24 ft, smooth landing 30 ft away.

8. Write $h(x) = a(x - 30)^2(x + 15)$ and find a.

9. Expand your polynomial to standard form.

10. Use synthetic division to verify a height at $x = 10$.

Final Product: Slide Design Report

Your final product must include:

- Both forms of each slide
- Theorem checks
- End behavior and graph
- Design report

Reflection

R1. How do the Factor and Remainder Theorems save time?

R2. Why does the factor (x + 20) matter even though x = −20 is off the slide?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Original slide polynomial starts at the required height and lands smoothly (double zero), with work shown. Goes beyond the minimum with a creative, realistic choice.	Original slide polynomial starts at the required height and lands smoothly (double zero), with work shown.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Report includes the polynomial in factored and standard form, a graph with zeros and multiplicity, division checks, and a design. Polished, organized, and visually engaging.	Report includes the polynomial in factored and standard form, a graph with zeros and multiplicity, division checks, and a design. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.